NYC 311 Service Requests

Big Data Analysis Using Hadoop, Hive, and Excel 3D Map

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CIS 4560: Introduction to Big Data
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Agenda

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2. Dataset Overview and Data Source Requirements
3. Hardware / Cluster Specifications
4. Hadoop, HDFS, Hive, and Beeline Workflow
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Project Objective

Analyze NYC 311 service requests from 2020 to present.

Identify common complaint types and request hotspots to prioritize high-volume city service issues.

Use location and time fields to support tempo-spatial analysis.

Process a large dataset with Hadoop/HDFS and Hive.

Create reproducible slides, paper, tutorial, and GitHub deliverables.

Dataset Overview and Data Source

Dataset Details

File: 311_Service_Requests_from_2020_to_Present.csv

Dataset size: 12.63 GB

Subset used for visualization: 10,000 records

Columns: 44

Time range: 2020 to present

Required Links

Data source: Kaggle dataset package

URL: https://www.kaggle.com/datasets/aliafzal9323/los-angeles-crime-data-2020-2026

GitHub: https://github.com/edwaaardgee/311-service-request-hadoop-analysis

Note: Kaggle package title differs, but the file used for this project is the NYC 311 Service Requests CSV.

Hardware / Cluster Specifications

Cluster Environment

Big Data platform: Hadoop / Hive

Cluster version: Hadoop 3.4.1

Storage: HDFS

Query interface: Beeline

Processing: HiveQL and YARN

Experimental Specs

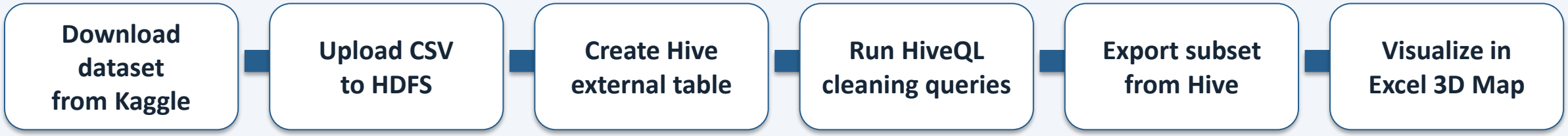
Cluster nodes: 5 nodes — 2 master, 3 worker

Memory: 31 GB RAM

CPU: AMD EPYC 7J13 processor

CPU speed: 2.4 GHz

Implementation Workflow / Architecture



This workflow shows the full implementation from raw data to processed visualization output.

Data Cleaning and Preparation

Removed records with missing or invalid latitude and longitude values.

Used created_date fields for time-based analysis in Excel 3D Map.

Focused on service request categories related to infrastructure, sanitation, and public services.

Selected relevant fields: date, agency, complaint type, borough, latitude, and longitude.

Exported a 10,000-record subset for Excel 3D Map due to visualization limits.

Hive Table Creation and Processing

Connected to the Hadoop cluster remotely using SSH.

Created an HDFS project directory for the 311 service request data.

Uploaded the large CSV file into HDFS.

Opened Beeline and selected the Hive database.

Created an external Hive table using OpenCSVSerde for CSV parsing.

Used HiveQL to preview records, count rows, filter fields, and prepare output.

HiveQL Analysis Performed

Main Query Goals

Count total records in the Hive table

Find top service request categories

Compare requests by location and category

Filter rows with valid latitude and longitude

Export a cleaned mapping dataset

HiveQL

```
SELECT problem, COUNT(*) AS total_requests
FROM service_requests_311
GROUP BY problem
ORDER BY total_requests DESC
LIMIT 10;
```

Purpose: Ranks service request categories by total count.

Service Request Category Analysis

Complaint-Type Analysis

Aggregated request totals by complaint/problem type using HiveQL.

Identified the highest-volume categories: Noise - Residential, Heat/Hot Water, Bulky Item Collection, Illegal Parking, and Blocked Driveway.

Used results to compare which city issues appeared most often in the dataset.

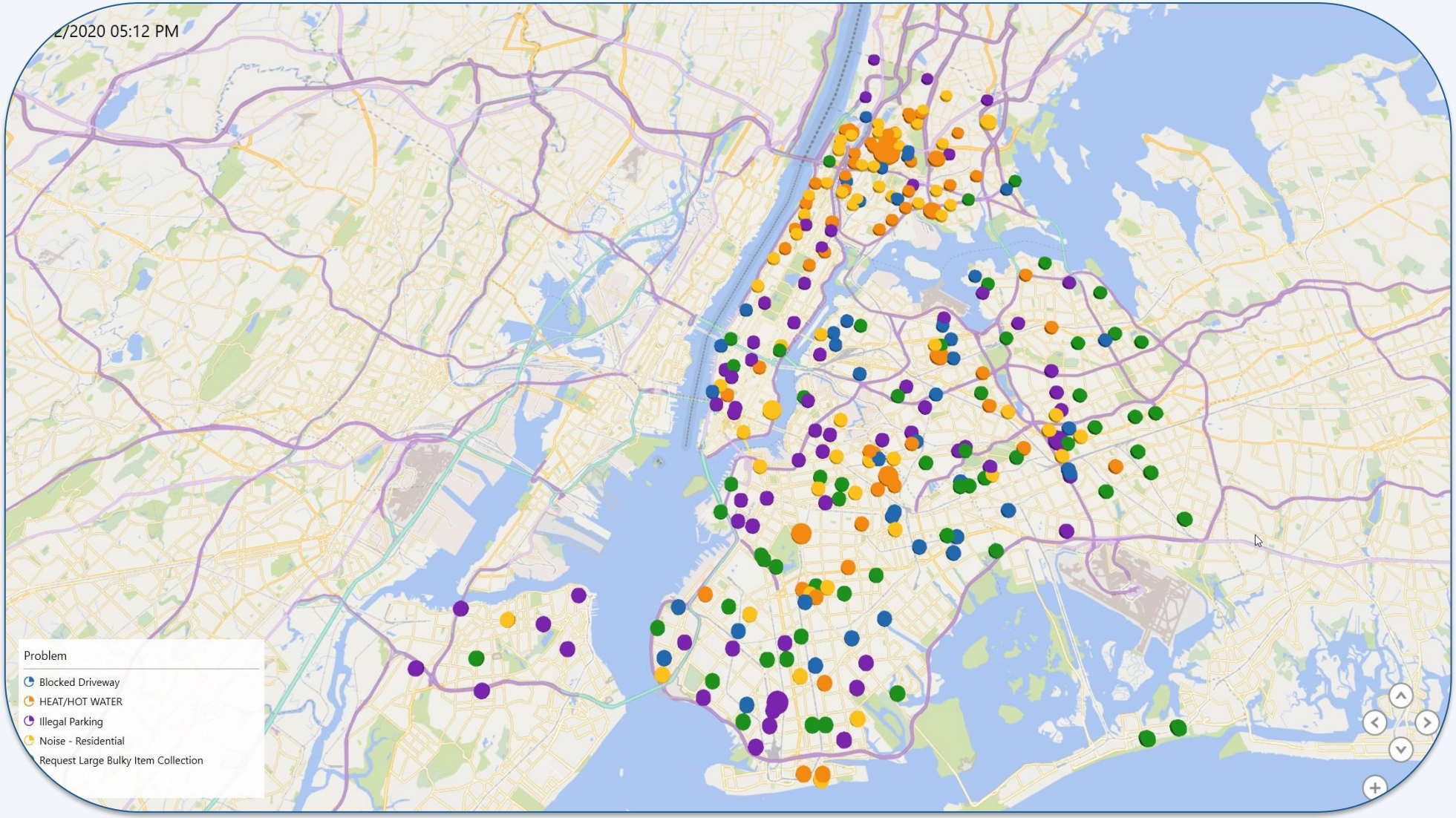
This analysis supports the recommendation to prioritize high-frequency complaints.

problem	total_requests
Noise - Residential	40629
HEAT/HOT WATER	35116
Request Large Bulky Item Collection	30512
Illegal Parking	24154
Blocked Driveway	14994
Consumer Complaint	13840
Street Condition	11480
Street Light Condition	10750
Non-Emergency Police Matter	9732
Noise - Street/Sidewalk	8696

rows selected (18.675 seconds)

Tempo-Spatial Visualization

What the Visualization Shows



Location fields used: latitude, longitude, borough, and incident location.

Time fields used: created date and service request timestamp.

Excel 3D Map visualized service requests by location and time.

Goal: show where requests cluster and how request patterns change over time.

Key Findings

Noise - Residential, Heat/Hot Water, and Bulky Item Collection were the top three service request categories based on the HiveQL count query.

High-volume complaints show which city issues should receive the most attention and resources.

Infrastructure and quality-of-life issues appeared repeatedly, including blocked driveways, street conditions, and street light problems.

Combining created_date with latitude and longitude revealed patterns that a basic table would not show

Recommendations

Prioritize the highest-volume complaint categories first, especially Noise - Residential, Heat/Hot Water, Bulky Item Collection, Illegal Parking, and Blocked Driveway.

Use hotspot locations from the 3D Map to guide staffing, field operations, and resource allocation.

Track repeated complaints over time to identify recurring neighborhood service problems.

Use Hive and HDFS to process large public-service datasets more efficiently than local tools.

Build geospatial dashboards so agencies can monitor service request trends and make faster data-driven decisions.

Thank You

Questions?